

Plan 2 in plain English

A companion to the second proof of the sharp quantitative Faber–Krahn inequality

Reading time ≈ 30 minutes. The aim is to convey what the proof is doing and why, not to replicate any estimate.

0. What is the problem?

The **Faber–Krahn inequality** says: among all open sets in $\mathbb{R}^n$ of a given volume, the ball minimises the first Dirichlet eigenvalue of the Laplacian:

$$\lambda_1(\Omega) \geq \lambda_1(B^*),$$

where B^* is the ball of the same volume as Ω . The classical proof (Faber 1923, Krahn 1925) uses symmetrisation.

We care about *how rigid* this is. If Ω is only “a little” non-ball — quantified by the **Fraenkel asymmetry**

$$\mathcal{A}(\Omega) = \inf_{x \in \mathbb{R}^n} \frac{|\Omega \Delta B^*(x)|}{|\Omega|}$$

(the symmetric difference between Ω and its best-matching ball) — then λ_1 should be only a little above $\lambda_1(B^*)$. The **sharp quantitative Faber–Krahn inequality** says exactly that:

$$\delta_{FK}(\Omega) \geq c(n) \mathcal{A}(\Omega)^2,$$

where $\delta_{FK}(\Omega) = (|\Omega|/\omega_n)^{2/n}(\lambda_1(\Omega) - \lambda_1(B^*))$ is the normalised eigenvalue gap. The exponent 2 is *sharp* (an ellipsoid with one axis stretched by ε gives $\mathcal{A} \sim \varepsilon$ and $\delta_{FK} \sim \varepsilon^2$, so you can’t ask for any smaller exponent on the right). Proving this was the conjecture closed by Brasco–De Philippis–Velichkov (BDV) in 2015. Plan 1 of this manuscript polishes their argument; Plan 2 gives a *different* proof.

Two simplifications before we start

Trade λ_1 for the torsion function. It is much easier to work with the **torsion function** u_Ω , the unique solution of

$$-\Delta u = 1 \text{ in } \Omega, \quad u = 0 \text{ on } \partial\Omega.$$

The **torsional rigidity** is $T(\Omega) = \int_\Omega u$. Saint–Venant gives $T(\Omega) \leq T(B^*)$ and a corresponding deficit $\delta_T(\Omega)$.

The two stability problems are *not independent*. Kohler–Jobin (1978) proved

$$T(\Omega) \lambda_1(\Omega)^{(n+2)/2} \geq T(B^*) \lambda_1(B^*)^{(n+2)/2},$$

and a small computation turns this into:

sharp δ_T-stability $\implies$ sharp δ_{FK}-stability

with an explicit dimensional constant. **So our entire job reduces to proving $\delta_T(\Omega) \geq c(n) \mathcal{A}(\Omega)^2$.** That is the target. Plan 1 and Plan 2 both produce this same input.

Reduce to bounded Ω . A separate lemma (BDV, Lemma 5.3) shows that for stability you may assume $\Omega \subset B_R$ for some explicit $R = R(n)$. Constants in the proof may depend on R ; bounded reduction at the end removes that dependence. This is just bookkeeping, common to both plans.

So: **the problem is $\delta_T(\Omega) \geq c(n, R) \mathcal{A}(\Omega)^2$ for bounded Ω .**

1. The big idea of Plan 2

Plan 1 (BDV) is a **selection** argument: find one near-extremal Ω , prove it is smooth enough to be a graph over the unit sphere, then quote Fuglede’s *nearly spherical theorem*, which says a near-ball that is actually a graph satisfies the inequality directly.

Plan 2 is a **foliation** argument. Instead of looking at Ω as a single shape, slice u_Ω into its superlevel sets

$$E_\rho := \{u_\Omega > t(\rho)\}, \quad |E_\rho| = \omega_n \rho^n.$$

As ρ runs from 0 (top of u) to 1 (whole Ω), the sets E_ρ swell from a point up to Ω . Each is a finite-perimeter set. By Talenti’s classical theorem, the rearrangement of u_Ω matches the rearrangement of the torsion function of B^* , so the *volumes* behave correctly; the question is how the *shapes* behave.

The strategy:

Watch every level set, control it using Fusco–Julin isoperimetric stability, integrate against a clever weight, and read off control of $\mathcal{A}(\Omega)$ at the end.

Six things have to happen in sequence:

1. **The Saint–Venant deficit must be expressible as a ρ -integral** over the foliation. (level-set deficit identity)
2. **The foliation must be a well-defined path in some metric space of shapes**, so that we can talk about its “speed” and its “distance from the unit ball”. (metric framework)
3. **Each level set must have a centre** with respect to which we can quantify how non-ball it is. (Fusco–Julin centres)
4. **A geometric estimate (the radial trace lemma) must convert “distance from ball” to defects the level set already controls.** (oriented L^1 radial trace)
5. **Integrating along the foliation must produce a single special radius $\hat{\rho}$ where $F_{\hat{\rho}}$ is genuinely close to B_1 .** (weighted metric trace)
6. **That control must transfer back to Ω itself**, with the boundary layer $\Omega \setminus E_{\hat{\rho}}$ behaving tamely. (boundary-layer transfer)

Each is a separate idea, each is necessary, and none is conditional.

2. Step 1 — The level-set deficit identity

This is the **foundation**. There is a non-negative weight $w(\rho)$ on $(0, 1)$ and two non-negative defects $D_H(t)$, $D_I(t)$ such that

$$\delta_T(\Omega) \simeq \int_0^1 (D_H(t(\rho)) + D_I(t(\rho))) w(\rho) d\rho.$$

The deficit, an a-priori opaque single number, is *exactly* the integral of two pointwise non-negative defects along the foliation. **A single global inequality has become an integrated identity.**

What do D_H and D_I measure?

Both come from rewriting Saint–Venant’s proof one level at a time. Define

$$\alpha(t) = \int_{\partial^* E_t} \frac{d\mathcal{H}^{n-1}}{|\nabla u|}, \quad \beta(t) = \int_{\partial^* E_t} |\nabla u| d\mathcal{H}^{n-1}.$$

By Cauchy–Schwarz, $\alpha(t)\beta(t) \geq P(E_t)^2$, with equality iff $|\nabla u|$ is constant on $\partial^* E_t$.

- $D_H(t) = \alpha(t)\beta(t) - P(E_t)^2 \geq 0$ — the **Cauchy–Schwarz defect**. Measures oscillation of $|\nabla u|$ along the level set. Zero when $|\nabla u|$ is constant on $\partial^* E_t$. (H for “Hessian-like”.)
- $D_I(t) = P(E_t)^2 - c_n |E_t|^{2-2/n} \geq 0$ — the **squared isoperimetric defect** of the level set. Zero when E_t is a ball. (I for “Isoperimetric”.)

So the identity says: the Saint–Venant deficit is the integral over levels of (gradient oscillation on the level) + (how far the level is from a ball), weighted by something explicit.

Why this is true

Talenti’s rearrangement proof of $T(\Omega) \leq T(B^*)$ involves two Cauchy–Schwarz-like steps — one giving an isoperimetric inequality on level perimeters, one on gradient integrals. Each contributes a non-negative defect at every level. Plan 2’s first move is to refuse to throw these defects away. Saint–Venant becomes an *equality* in which the two defects sum, integrated against an explicit weight, to give δ_T .

This is in the spirit of Nicola–Tilli’s recent work on the short-time Fourier Faber–Krahn problem. Plan 2 imports the idea into the Dirichlet Saint–Venant world.

Why it matters

Two non-negative summands on the right that sum to something small means that *both* are integrally small:

$$\int D_H(t(\rho))w(\rho) d\rho \leq \delta_T, \quad \int D_I(t(\rho))w(\rho) d\rho \leq \delta_T.$$

The first will control the *speed* of the foliation (Step 2/4); the second will control how *ball-shaped* each level is (Step 3).

3. Step 2 — The metric framework

We want to think of the family $\{E_\rho\}$ as a **path** in a space of shapes. Then “speed” and “distance from a target” make sense, and standard metric-space arc-length tools apply.

The space $\mathcal{X}$

Define $\mathcal{X}$ to be the space of finite-perimeter sets in $\mathbb{R}^n$ *modulo translations*, with distance

$$d_{\mathcal{X}}([A], [B]) = \inf_{v \in \mathbb{R}^n} |A \Delta (B + v)|.$$

Two sets are at distance zero iff they’re translates of one another. The completion is Polish (“nice enough” that Ambrosio–Gigli–Savaré metric arc-length theory works).

The path

Consider the **rescaled** foliation

$$F_\rho := \rho^{-1} E_\rho.$$

Why rescale? Because $|E_\rho| = \omega_n \rho^n$, so $|F_\rho| = \omega_n = |B_1|$ for every ρ . All the F_ρ live on the same volume scale, so it is meaningful to compare them to the unit ball B_1 .

Plan 2 proves two things about this path:

- **It is absolutely continuous.** $\rho \mapsto [F_\rho]$ has finite metric derivative $|\dot{F}_\rho|_{\mathcal{X}}$ a.e.
- **The metric speed is bounded geometrically.** Roughly, $|\dot{F}_\rho|_{\mathcal{X}}$ is controlled by the L^1 norm of the normal velocity $V_\rho = -t'(\rho)/|\nabla u|$ on $\partial^* E_\rho$, after rescaling.

This step is *almost smooth-flow geometry*. If Ω were smooth and u_Ω Morse, the level sets would move with normal velocity V_ρ and a one-line first-variation argument would give the bound. The work is to make this rigorous when Ω is only of finite perimeter — no smoothness, no Lipschitz boundary. One mollifies u_Ω , applies the smooth identity to the mollified levels, and passes to the limit using Reshetnyak lower-semicontinuity. The technical care is non-trivial but the **idea** is just: the foliation is a Lipschitz path, in the right metric.

4. Step 3 — Fusco–Julin centres and the oriented radial trace

This is the heart of Plan 2. We must convert isoperimetric defect into F_ρ 's distance from the unit ball.

The Fusco–Julin inequality

The classical Fuglede–Fusco–Maggi–Pratelli inequality (2008) says: for any E with the volume of a ball,

$$\mathcal{A}(E)^2 \leq C(n) \mathcal{D}(E),$$

where $\mathcal{D}(E) = (P(E) - P(B^*)) / P(B^*)$ is the linear isoperimetric deficit. Exponent 2 is sharp.

Fusco–Julin (2014) strengthened this to a *two-term* inequality:

$$\mathcal{A}(E)^2 + \beta(E)^2 \leq C(n) \mathcal{D}(E), \quad \beta(E)^2 = \inf_{z \in \mathbb{R}^n} \int_{\partial^* E} \left| \nu_E(x) - \frac{x - z}{|x - z|} \right|^2 d\mathcal{H}^{n-1}.$$

For the optimal centre z , $\beta(E)$ measures how much the outer normal ν_E differs from the radial direction pointing away from z — i.e., how much $\partial^* E$ fails to be a sphere around z .

Both terms — symmetric difference *and* normal oscillation — are controlled by the (single) isoperimetric deficit. **This is what makes a foliation argument possible:** at every level, FJ gives us two pieces of geometric information, and we need both.

The Fusco–Julin centres

For each ρ where $D_I(t(\rho))$ is small enough — the **good levels** — pick a centre z_ρ that almost realises the FJ infimum. The canonical concrete choice is the **bulk centroid** of E_ρ ; it's automatically Borel in ρ and (with a small Wave 3 argument) it sits well inside E_ρ , which is exactly the analytic input we need below.

So now we have, for a.e. good ρ :

- (asymmetry) $|E_\rho \Delta B_\rho(z_\rho)| \leq \rho^n \cdot O(\sqrt{D_I(t(\rho))})$,
- (oscillation) $\int_{\partial^* E_\rho} |\nu_\rho - (x - z_\rho)/|x - z_\rho||^2 d\mathcal{H}^{n-1} \leq O(D_I(t(\rho)))$.

The oriented L^1 radial trace lemma

This is the most delicate step and the **load-bearing original lemma** of Plan 2:

$$\boxed{\int_{\partial^* E_\rho} \left| \frac{|x - z_\rho|}{\rho} - 1 \right| d\mathcal{H}^{n-1} \leq C \left(\underbrace{|E_\rho \triangle B_\rho(z_\rho)|}_{\text{symmetric difference}} + \underbrace{\int_{\partial^* E_\rho} \left| \nu_\rho - \frac{x - z_\rho}{|x - z_\rho|} \right|^2 d\mathcal{H}^{n-1}}_{\text{normal oscillation}} \right)}.$$

In words: the **average distance of $\partial^* E_\rho$ from the sphere of radius ρ around z_ρ** is controlled by the FJ defects.

The intuition. Think of $\partial^* E_\rho$ as a closed surface near the sphere of radius ρ centered at z_ρ . The integrand $||x - z_\rho|/\rho - 1|$ measures, at each boundary point, how far its distance to z_ρ deviates from ρ . The left side is the total “radial deviation” of the surface.

We control it by **applying the divergence theorem to the radial vector field**

$$X(x) = \left| \frac{|x - z_\rho|}{\rho} - 1 \right| \cdot \frac{x - z_\rho}{|x - z_\rho|}.$$

Outside the sphere $|x - z_\rho| = \rho$ this points outward radially; inside, inward radially. The divergence theorem says

$$\int_{\partial^* E_\rho} X \cdot \nu_\rho d\mathcal{H}^{n-1} = \int_{E_\rho} \operatorname{div} X.$$

The right side expands into a *signed* version of $|E_\rho \triangle B_\rho(z_\rho)|$ plus error. The left side bounds the radial trace only if you replace $X \cdot \nu_\rho$ by $|X|$, which costs an oscillation term $|\nu_\rho - (x - z_\rho)/|x - z_\rho||$. Squaring gives the second term on the right. Combining: the inequality.

Why this is delicate.

1. The vector field X has a **singularity at z_ρ** . You can’t apply the divergence theorem naively — you must truncate X near z_ρ , apply the theorem to a smooth truncation, and pass to the limit. This is the *truncation argument*, made rigorous for finite-perimeter sets in the preliminaries.
2. For the truncation to be free, z_ρ must be **interior** to E_ρ at uniform positive distance. This is exactly why we chose the bulk centroid: it satisfies $B(z_\rho, \rho_*/4) \subset E_\rho$ on good levels. The symmetric-difference region then stays away from z_ρ and the truncation limit is *trivial* (not just convergent).
3. The lemma is *linear* in the radial deviation (left side) and combines a linear term plus a *squared* oscillation term on the right. **This linear-vs-squared mismatch is what makes the final rate sharp instead of degraded.**

Why this is the load-bearing step. Earlier attempts tried to use L^2 radial traces (square the left side too) and *ray-slicing* arguments. Both failed: Wave 2 audits flagged the L^2 version as inconsistent with the FJ scaling, and ray-slicing doesn’t work for finite-perimeter sets. **The L^1 oriented trace via divergence theorem with truncation is what saves Plan 2.** The audit (Agent 16) explicitly verifies that the named vector field X , the truncation argument, and the finite-perimeter divergence theorem are all present.

5. Step 4 — The weighted metric trace

We now have, for each good level:

- a centre z_ρ ,
- a radial trace estimate (Step 3),
- the level-set identity for δ_T (Step 1),
- the metric derivative bound for the path F_ρ (Step 2).

We want to find **one specific $\hat{\rho}$ near 1** such that $F_{\hat{\rho}}$ is close to B_1 in $\mathcal{X}$.

The integrated action inequality

Chaining the above estimates: the radial trace, rescaled, turns the FJ pair (asymmetry + oscillation) into a single bound on $d_{\mathcal{X}}([F_\rho], [B_1])^2$ at each good level. Combining with the metric derivative bound (Step 2, controlled by D_H via the normal-velocity formula), and integrating against the natural weight $\mu(d\rho) = -t'(\rho) d\rho$:

$$\int_{[\rho_*, 1]} \left(d_{\mathcal{X}}([F_\rho], [B_1])^2 + |\dot{F}_\rho|_{\mathcal{X}}^2 \right) \mu(d\rho) \leq C(n, R, \rho_*) \delta_T(\Omega).$$

This is the metric-space avatar of the level-set deficit identity. The integrated “action” of the path F_ρ (distance to target + speed squared, weighted) is controlled by the deficit.

Good vs. bad levels

A subtle point: FJ only gives information on levels where D_I is small. On *bad* levels (where D_I is above some fixed threshold), we have no centre, no radial trace, nothing. **Markov’s inequality saves the day**: the bad-level set has weighted measure at most $\delta_T/\text{threshold}$, so it’s a tiny piece of $[\rho_*, 1]$ when δ_T is small.

A separate good/bad decomposition is needed for the levels where the *profile slope* $-t'(\rho)$ is small. There the weight μ is too small for the integrated bound to be useful. Talenti’s bad-set Lebesgue estimate gives that this bad- μ region has Lebesgue measure $O(\delta_T)$, and a uniform Lipschitz bound on the path absorbs it without rate loss.

After both decompositions, the integrated action holds on the *whole* interval $[\rho_*, 1]$, with the bad-set contributions absorbed.

The abstract metric trace argument

We have an absolutely continuous path $\rho \mapsto [F_\rho]$ in $\mathcal{X}$, an integrated bound on (distance to B_1)² + speed², and we want to extract **one good radius $\hat{\rho}$** .

This is a classical arc-length argument. Heuristically: if $\int (d^2 + |\dot{F}|^2) d\mu \leq \varepsilon$ over a window of weighted length L^* , then by averaging there is some $\hat{\rho}$ in the window where $d([F_{\hat{\rho}}], [B_1])^2 \leq \varepsilon/L^*$ (Markov in d^2) and the kinetic transport within the window is bounded by $L^* \cdot (\text{speed term})$.

The critical choice — and this is exactly what the audit (B1) caught and the repair fixed — is the **window scale**:

$$\hat{\rho} \in [\rho_\delta - c_0 \sqrt{\delta_T}, \rho_\delta], \quad \rho_\delta = 1 - \kappa \sqrt{\delta_T}.$$

The window has length $c_0 \sqrt{\delta_T}$, *not* a fixed $O(1)$ length. With this scale the boundary layer $\Omega \setminus E_{\hat{\rho}}$ will be $O(\sqrt{\delta_T})$ in measure (Step 5). The original draft had an $O(1)$ window, which would have *broken* Step 5; the repair fixed it.

The output:

$$\boxed{d_{\mathcal{X}}([F_{\hat{\rho}}], [B_1])^2 \leq C_*(n, R, \rho_*) \delta_T(\Omega).}$$

In words: there is a level $\hat{\rho}$ very close to the top of the foliation at which the rescaled superlevel set is within distance $\sqrt{\delta_T}$ of a unit ball in the shape metric.

6. Step 5 — Boundary-layer transfer and assembly

We have $F_{\hat{\rho}}$ close to B_1 in $\mathcal{X}$. We want Ω close to B^* . There is a gap: $E_{\hat{\rho}}$ is a *superlevel set* of Ω , not Ω itself.

The boundary layer

Since $\hat{\rho} \in [\rho_\delta - c_0\sqrt{\delta_T}, \rho_\delta]$ and $|E_\rho| = \omega_n \rho^n$,

$$|\Omega \setminus E_{\hat{\rho}}| = \omega_n(1 - \hat{\rho}^n) \leq C\sqrt{\delta_T(\Omega)}.$$

This *is* a rate loss — “ $E_{\hat{\rho}}$ differs from Ω by $O(\sqrt{\delta_T})$ ” — but it’s harmless if handled correctly.

The squaring trick

We have:

- $\mathcal{A}(E_{\hat{\rho}}) \leq O(\sqrt{\delta_T})$ (after unscaling from $F_{\hat{\rho}}$ to $E_{\hat{\rho}}$).
- $\mathcal{A}(\Omega) \leq \mathcal{A}(E_{\hat{\rho}}) + 2|\Omega \setminus E_{\hat{\rho}}|/|\Omega| \leq \mathcal{A}(E_{\hat{\rho}}) + C\sqrt{\delta_T}$ (triangle inequality for asymmetry).

So $\mathcal{A}(\Omega) \leq O(\sqrt{\delta_T})$. **Squaring** gives $\mathcal{A}(\Omega)^2 \leq O(\delta_T)$ — the sharp exponent. Concretely, $(a + b)^2 \leq 2a^2 + 2b^2$ doubles a constant but does *not* degrade the exponent.

This is the famous “**square at the end, not in the middle**” principle. Earlier drafts of Plan 2 (the centroid / Π -route, now an auxiliary remark) squared too early and ended with $\sqrt{\delta_T}$ on the right of the final inequality — the wrong rate. The repaired Plan 2 squares *after* the transfer to Ω and preserves exponent 2.

Bounded reduction and Kohler–Jobin

At this point:

$$\mathcal{A}(\Omega)^2 \leq C(n, R) \delta_T(\Omega) \quad \text{for } \Omega \subset B_R.$$

Bounded reduction (BDV Lemma 5.3, imported from Plan 1’s `BoundedReduction` block) removes the R dependence; **Kohler–Jobin** transfers δ_T -stability to δ_{FK} -stability. Both reductions are common to both Plans, so Plan 2’s final theorem

$$\delta_{FK}(\Omega) \geq c_{FK}(n) \mathcal{A}(\Omega)^2$$

is stated in *the same normalisation and with the same dimensional constant* as Plan 1’s.

7. A one-page summary

Read this when you forget what just happened.

$\delta_T(\Omega)$
↓ <i>Step 1: level-set deficit identity</i>
$\int [D_H(t(\rho)) + D_I(t(\rho))] w(\rho) d\rho$
↑ gradient osc. ↑ isoperimetric defect
↓ <i>Step 2: rescale $E_\rho \rightarrow F_\rho$; metric framework</i>
↓ controls $ \dot{F}_\rho _{\mathcal{X}}$ by D_H
$\int \dot{F}_\rho _{\mathcal{X}}^2 d\mu \leq (D_H \text{ term}) \leq \delta_T$
↓ <i>Step 3: Fusco--Julin centre z_ρ</i>
↓ + <i>ORIENTED RADIAL TRACE LEMMA</i>
↓ converts D_I into $d_{\mathcal{X}}(F_\rho, B_1)^2$
$\int [d_{\mathcal{X}}([F_\rho], [B_1])^2 + \dot{F}_\rho _{\mathcal{X}}^2] d\mu \leq C \cdot \delta_T$
↓ <i>Step 4: weighted metric trace</i>
↓ on window $[\rho_\delta - c_0\sqrt{\delta_T}, \rho_\delta]$
$\exists \hat{\rho}: d_{\mathcal{X}}([F_{\hat{\rho}}], [B_1])^2 \leq C_* \cdot \delta_T$
↓ <i>Step 5: $\Omega \setminus E_{\hat{\rho}} = O(\sqrt{\delta_T})$</i>
↓ <i>transfer + squaring trick</i>
$\mathcal{A}(\Omega)^2 \leq C(n, R) \cdot \delta_T(\Omega)$
↓ <i>Bounded reduction (removes R)</i>
↓ <i>Kohler--Jobin ($\delta_T \rightarrow \delta_{FK}$)</i>
$\delta_{FK}(\Omega) \geq c_{FK}(n) \cdot \mathcal{A}(\Omega)^2$

8. How Plan 2 differs from Plan 1

Aspect	Plan 1 (BDV)	Plan 2 (level-set/metric)
Object	One near-extremal Ω found by selection	The whole foliation $\{E_\rho\}$ of u_Ω
Regularity used	$C^{2,\gamma}$ graph over the sphere	Finite perimeter only
Isoperimetric input	Fuglede's nearly-spherical theorem	Fusco–Julin two-term sharp inequality
Closing step	Contradiction at the selected minimiser	Arc-length trace argument in $\mathcal{X}$
Common final step	Bounded reduction + Kohler–Jobin	Same

Why bother including Plan 2?

1. **Structurally different.** It doesn't smooth Ω ; it doesn't pick out one near-ball; it doesn't graph anything over a sphere. It treats Ω as a foliation and watches every level.
2. **Direct use of FJ.** Plan 1 uses isoperimetric stability only indirectly, through regularity. Plan 2 plugs FJ into every level set; the role of quantitative isoperimetry is laid bare.
3. **Finite-perimeter native.** The whole proof works for finite-perimeter Ω with no graph assumption. The metric framework and the radial trace lemma are designed for finite perimeter from the start.
4. **A new lemma.** The oriented L^1 -radial trace lemma (Step 3) is, to our knowledge, new in this Saint–Venant / Faber–Krahn context. It is the conceptual core of the proof.

9. What is *not* used in Plan 2

The manuscript also contains an **auxiliary** section on the **centroid kinematic identity** and the **space-time Π -identity**. These are diagnostic computations:

- The bulk centroid $\bar{z}_\rho$ as a function of ρ has a kinematic formula (Lipschitz in ρ , derivative

computable from $|\nabla u|$ on the level set).

- A space-time identity decomposes the signed “ Π -moment” $\int |x - \bar{z}_\rho| - \rho$ on $\partial^* E_\rho$ across ρ .

These are *not* load-bearing for the repaired Plan 2. The original first attempt at Plan 2 tried to close the chain through pointwise Π -control instead of the oriented radial trace lemma, and required an *integrated* Π -control estimate we could not prove. The repaired chain (Steps 1–5) bypasses the Π -route entirely. The auxiliary section is kept as a record of the route and a place to upgrade if a future integrated Π -bound is proved, but **nothing in Plan 2 currently depends on it**. The auditor (Agent 16) verified this explicitly.

10. Where to look in the manuscript

If you want to read about...	Look at
Step 0 (Kohler–Jobin bridge, Talenti, FJ, GMT toolbox)	01_preliminaries_draft.tex
Step 1 (level-set identity)	plan2_levelset_identity.tex
Step 2 (metric framework, first variation)	plan2_metric_framework.tex
Step 3 (FJ centres, oriented radial trace)	plan2_fj_center_radial_trace.tex
Step 4 (weighted metric trace)	plan2_weighted_metric_trace.tex
Step 5 (boundary layer, assembly)	plan2_boundary_layer_assembly.tex
Auxiliary centroid/ Π -route (not load-bearing)	plan2_aux_centroid_note.tex
What the audit found and how it was repaired	audit_plan2.md, audit_plan2_repair_report.md
What’s the final verdict	final_adversarial_audit.md, audit_final_repair_report.md

The compiled manuscript is `Manuscript/main.pdf` (119 pages).

11. The two-sentence version

Plan 2 expresses the Saint–Venant deficit as an integral over the superlevel-set foliation of the torsion function, controls each level using Fusco–Julin centres and an oriented L^1 radial trace lemma, and extracts a single near-ball level via a weighted metric-space arc-length argument; transferring back to Ω and squaring at the end (not in the middle) gives the sharp $\mathcal{A}(\Omega)^2 \leq C \delta_T(\Omega)$. Kohler–Jobin then turns this into the sharp quantitative Faber–Krahn inequality with the same dimensional constant Plan 1 produces.